#B HKDF-Expand is used to derive the key from the master HKDF key

The Common IV is derived in almost the same way as the sender and recipient keys, as shown in listing 13.9. The label "IV" is used instead of "Key", and the length of the IV or nonce used by the COSE authenticated encryption algorithm is used instead of the key size. For example, the default algorithm is AES_CCM_16_64_128, which requires a 13-byte nonce, so you would pass 13 as the `ivLength` argument. Because our HKDF implementation returns a `Key` object, you can use the `getEncoded()` method to convert that into the raw bytes needed for the Common IV. Add this method to the `Oscore` class you just created.

Listing 13.9 Deriving the Common IV

```
private static byte[] deriveCommonIV(Key hkdfKey,
    byte[] idContext, AlgorithmID coseAlgorithm, int
ivLength)
        throws GeneralSecurityException {
    CBORObject context = CBORObject.NewArray();
    context.Add(new byte[0]);
    context.Add(idContext);
    context.Add(coseAlgorithm.AsCBOR());
    context.Add(CBORObject.FromObject("IV"));    #B
    context.Add(ivLength);    #B

    return HKDF.expand(hkdfKey, context.EncodeToBytes(),
#C
            ivLength, "dummy").getEncoded();    #C
}
```

#A Use the label "IV" and the length of the required nonce in bytes